



**UNIVERSITY OF
PLYMOUTH**

**School of Engineering, Computing, and
Mathematics**

PUSL3198

**Design and Control of Renewable Energy
Technology**

MPPT LAB Exercise Report

W.L.P. T. N. Wijayasinghe

10955873

5/22/2026

Abstract

This document outlines the design, construction, and assessment of a Maximum Power Point Tracking (MPPT) system that employs a DC-DC boost converter to enhance power extraction from a 10W photovoltaic (PV) array. The hardware design focuses on an IRF510 MOSFET controlled by a TC4429 isolated gate driver operating at a frequency of 100 kHz. During the construction phase, essential hardware optimizations were implemented, such as the incorporation of a custom 3C90 ferrite toroidal inductor to avoid magnetic saturation when handling high currents, along with the empirical adjustment of the gate resistance to 56 ohms to effectively dampen parasitic L-C oscillations. The system's telemetry for the ST ARM microcontroller was accurately implemented using a 5:1 voltage divider, a 1-ohm current shunt, and MCP602 operational amplifiers arranged with RC low-pass filters for signal conditioning. Theoretical power loss estimations derived from component datasheets indicate a highly efficient system operating at 94.65%, with diode conduction representing the primary loss vector. Finally, the Perturb and Observe (P&O) MPPT algorithm was implemented in C-code. A comparative analysis against Incremental Conductance (INC) and Fractional Open Circuit Voltage (FOCV) algorithms concludes that while P&O is highly effective for stable conditions, INC provides superior tracking reliability under rapidly varying solar irradiance.

Table of Contents

Abstract.....	i
Table of Contents	ii
List of figures	iii
Chapter 1 - Introduction	1
1.1 Introduction to Photovoltaic Energy Harvesting	1
1.2 The Maximum Power Point	1
1.3 Power Electronic Interfaces	1
1.4 Project Objectives	2
Chapter 2 – Design and Build	3
2.1 MOSFET Drive Circuit Assembly.....	3
2.2 Boost Converter Circuit assembly	7
2.3 Measurement and Protection Circuits	9
Chapter 3 – Power Loss Estimation	12
3.1 MOSFET Conduction Losses	12
3.2 Diode Conduction Losses	13
3.3 MOSFET Switching Losses	13
3.4 Magnetic Core Losses.....	14
Total System Efficiency	14
Chapter 4 – MPPT Algorithms	15
4.1 The Perturb and Observe (P&O) Algorithm	15
4.2 Alternative Algorithm 1: Incremental Conductance (INC).....	16
4.3 Alternative Algorithm 2: Fractional Open Circuit Voltage (FOCV)	18
4.4 Comparative Algorithmic Analysis	20
References	21
Appendices	22

List of figures

Figure 1:Schematic of the isolated MOSFET and gate driver testing circuit.....	3
Figure 2:Breadboard assembly of the IRF510 and gate driver	4
Figure 3:Oscilloscope capture illustrating a severely underdamped state.....	4
Figure 4: Oscilloscope capture illustrating an appropriately damped state.	5
Figure 5:Oscilloscope capture of the overall switching performance.	6
Figure 6:Schematic of the Boost Converter	7
Figure 7:Physical assembly of the optimized Boost Converter, featuring the toroidal inductor and parallel capacitor network.....	7
Figure 8:Scope trace of Continuous Conduction Mode (CCM).	8
Figure 9:implementation of the MCP602 signal conditioning	9
Figure 10:Scope verification of the voltage divider.....	10
Figure 11:Oscilloscope capture illustrating dynamic signal conditioning.	11
Figure 12: Flowchart of the P&O Algorithm	15
Figure 13: Flowchart of the Incremental Conductance Algorithm.	16
Figure 14 : Flowchart of the Fractional Open Circuit Voltage Algorithm	18

Chapter 1 - Introduction

1.1 Introduction to Photovoltaic Energy Harvesting

As the world shift towards renewable energy, optimizing renewable energy power systems has become the major focus because of the economic and sustainable improvement it can generate. Photovoltaic (PV) panels are an accessible source of clean energy; however, their overall efficiency depends on factors such as panel material, operating temperature, sunlight intensity, panel orientation, and environmental factors like shading and dust.

PV energy harvesting is the process of photon absorption from the sun directly into usable electrical energy using semiconductor materials. The electrical behavior of a PV system is defined by its current-voltage (I-V) and power-voltage (P-V) curves. These non-static curves constantly change due to overall efficiency factors we discussed.

1.2 The Maximum Power Point

The Maximum Power Point (MPP) is the most power generated maximum point but due to these overall efficiency factors, The Maximum Power point changes. An MPPT system dynamically adapts to them. Its primary goal is to adjust the electrical load so that it extracts at maximum efficiency and the system needs to constantly monitor the MPP.

1.3 Power Electronic Interfaces

Power electronic interfaces are the critical gap between PV arrays and the electrical grid or load. To solve the change in MPP, power electronic interfaces like DC-DC boost converters are utilized. These converters use a Maximum Power Point Tracking (MPPT) algorithm to actively adjust the circuit's switching behavior based on real-time voltage and current measurements by tweaking the duty cycle accordingly. The active feedback loop on the system forces the PV panel to operate at its absolute peak efficiency.

1.4 Project Objectives

This practical report focuses on power generation using a 10W, 20V Photovoltaic panel as the primary energy source. The main objective is to design a boost converter circuit, connect it to the PV cell, and adjust the parameters to extract the maximum power under varying conditions.

To achieve this, the report will specifically:

- Detail the assembly of the MOSFET drive and main Boost converter circuits.
- Analyze the analog voltage and current sensing circuits required for the micro-controller's ADC.
- Provide a comprehensive power loss estimation for the boost converter using component datasheets.
- Implement the Perturb and Observe (P&O) MPPT algorithm in C-code, and evaluate two alternative algorithms based on complexity, reliability, stability, and reaction time

Chapter 2 – Design and Build

2.1 MOSFET Drive Circuit Assembly

At 100kHz drive circuit was assembled using the IRF510 N-Channel MOSFET and a gate driver (TC4429). The ST ARM microcontroller's 3.3V output is insufficient to overcome the MOSFET's Miller plateau. This causes the low current source to slow transition and dissipate the power as heat loss. The driver resolves this by operating from a 12V supply and utilizing its exceptionally low internal output resistance (typically 2.1 ohm) to deliver up to 6A of peak current, charging the MOSFET instantaneously.

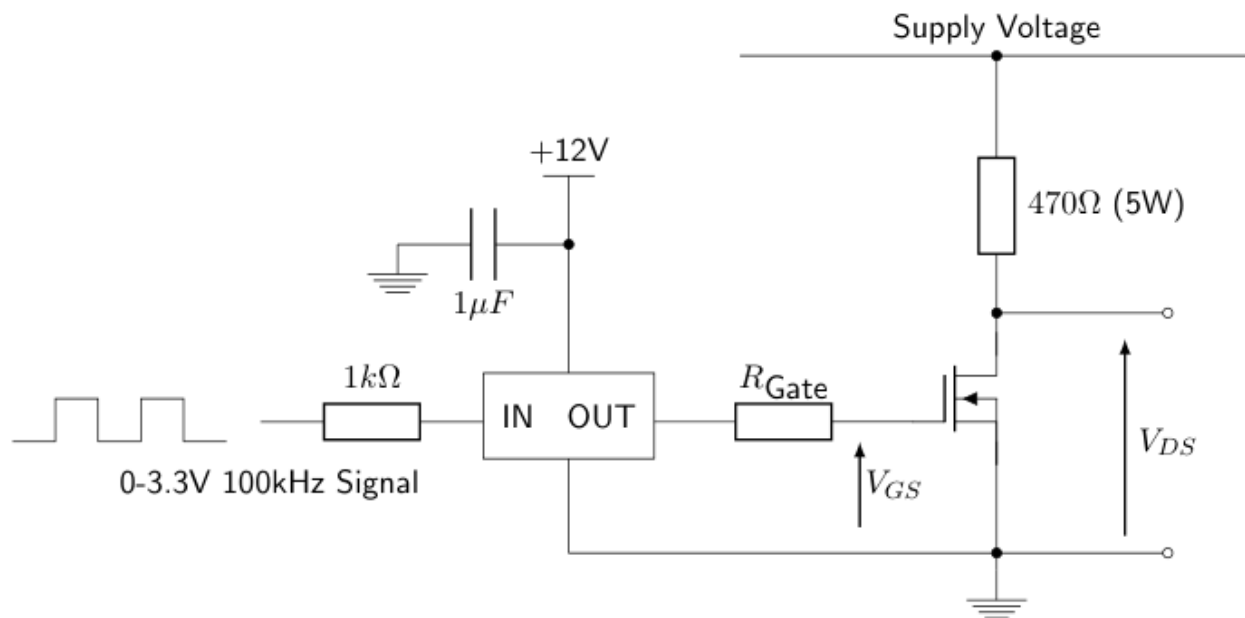


Figure 1: Schematic of the isolated MOSFET and gate driver testing circuit.

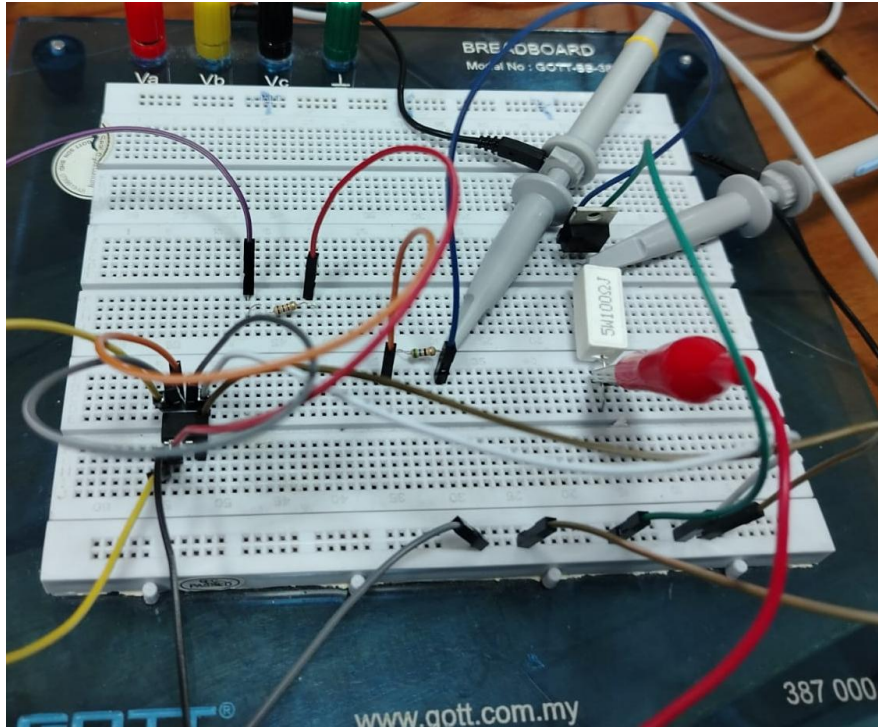


Figure 2: Breadboard assembly of the IRF510 and gate driver

Waveform Analysis and R_{Gate} Tuning:

The series gate resistor had to be tuned during testing. The driver's logic has an internal speed-up capacitor. If the output changes too fast, the high di/dt couples with parasitic breadboard trace inductance to form an underdamped L-C tank.



Figure 3: Oscilloscope capture illustrating a severely underdamped state.

As shown in Figure 3, testing with a 1-ohm resistor resulted in voltage overshoot and severe ringing on the rising edge of the gate signal. This high-frequency oscillation stresses the silicon gate dielectric and can lead to premature device failure.

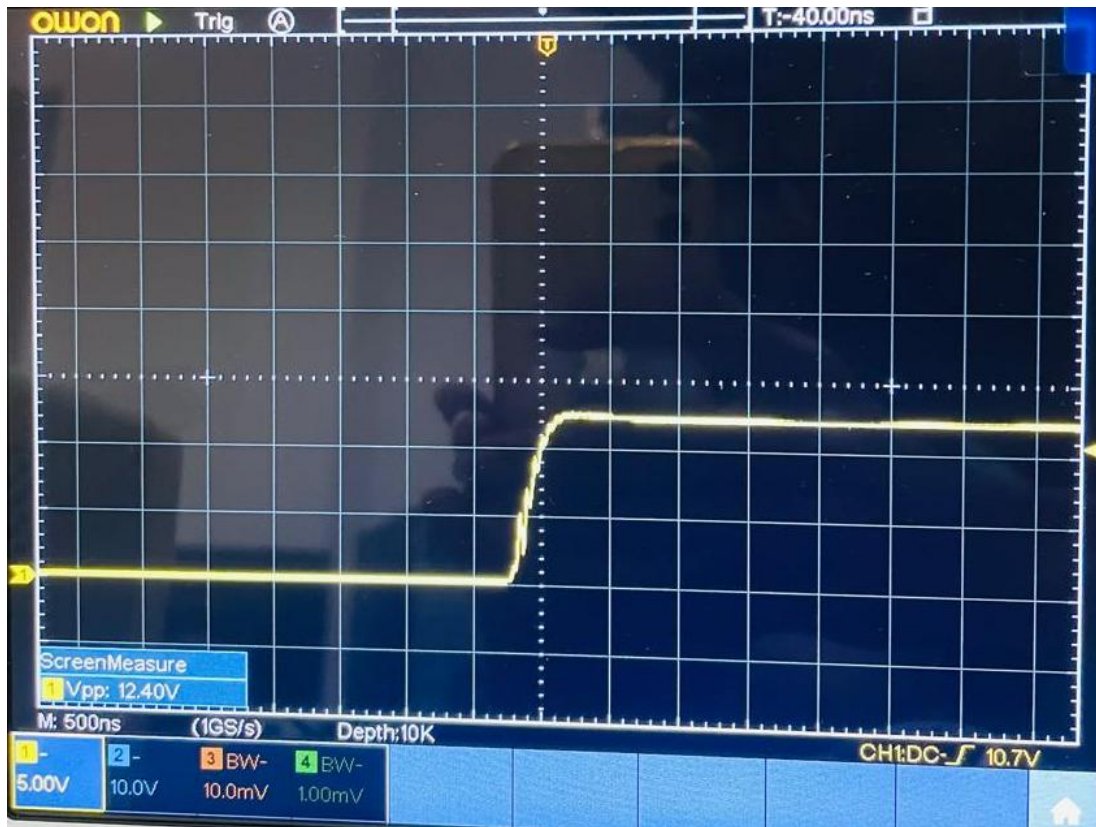


Figure 4: Oscilloscope capture illustrating an appropriately damped state.

To correct the underdamped response, the resistance was increased to 56 ohm. As demonstrated in Figure 4, this value provided sufficient damping to suppress the L-C tank oscillations.

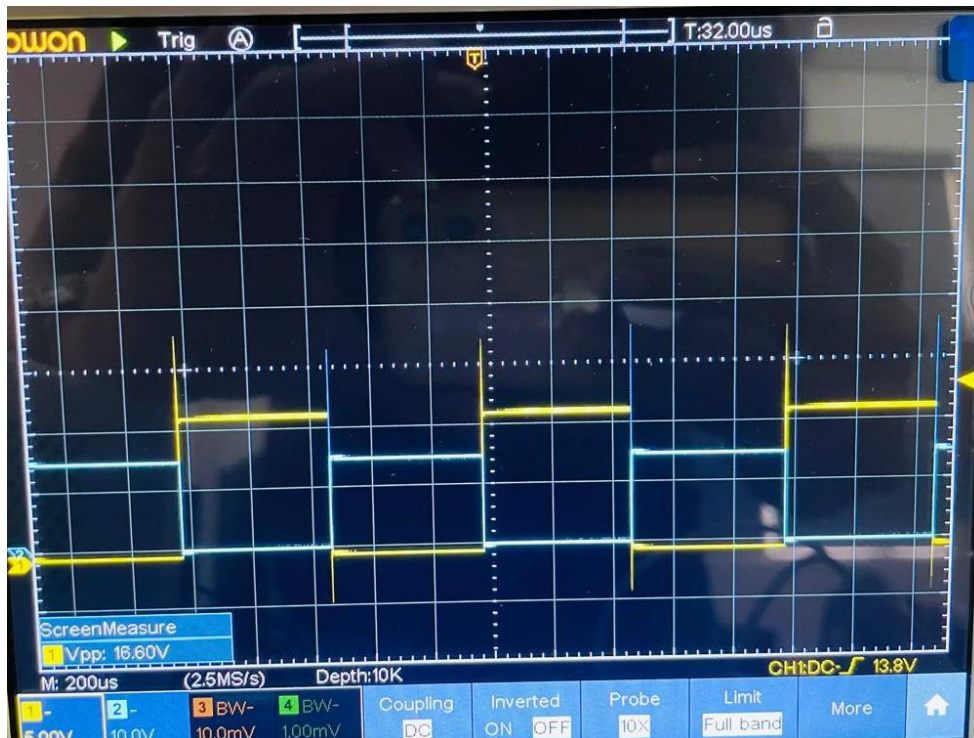


Figure 5: Oscilloscope capture of the overall switching performance.

Figure 5 confirms the overall operation of the drive circuit. The drain-to-source voltage cleanly and instantaneously inverts relative to the control signal, confirming that the MOSFET is achieving rapid saturation and cut-off with minimized switching losses.

2.2 Boost Converter Circuit assembly

The boost converter components were positioned around the MOSFET when the drive circuit was operating. To make sure the circuit could manage the 10W electricity from the PV panel throughout this test period, certain modifications to the lab manual instructions were required.

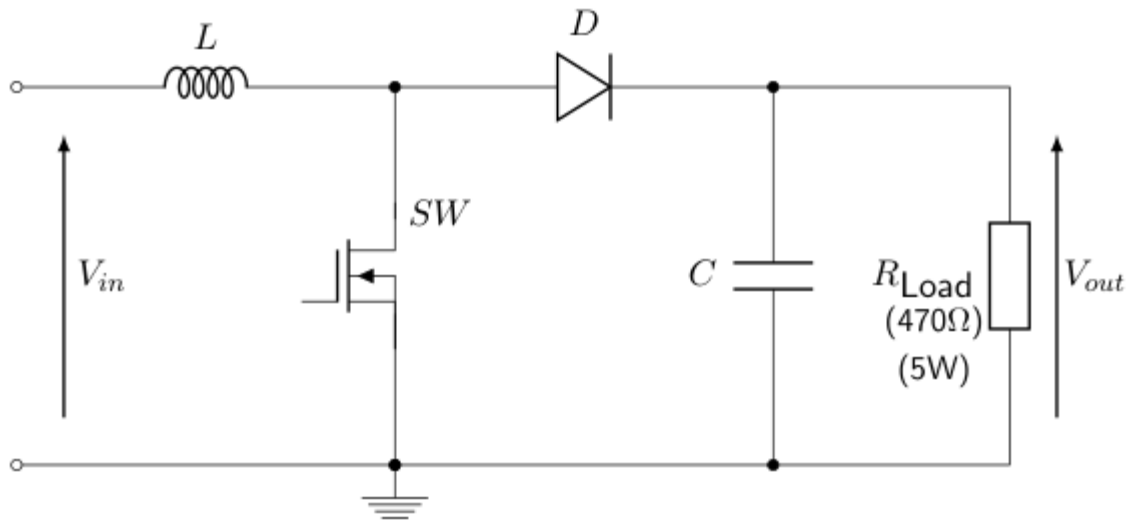


Figure 6: Schematic of the Boost Converter

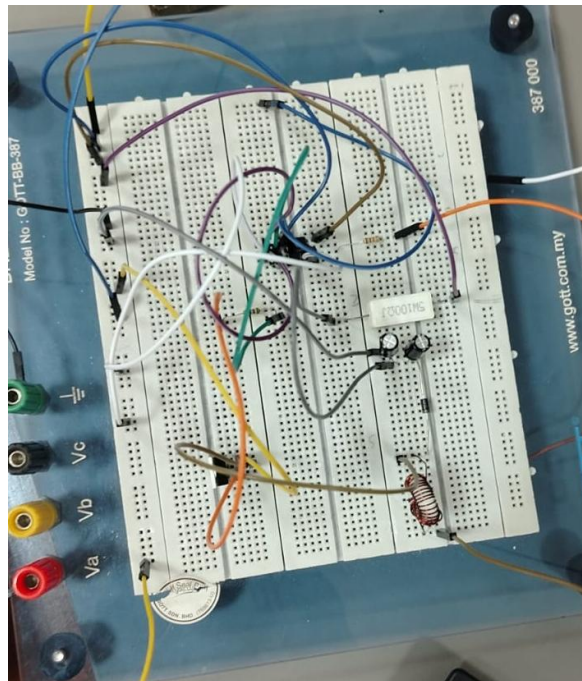


Figure 7: Physical assembly of the optimized Boost Converter, featuring the toroidal inductor and parallel capacitor network.

Modifications and Enhancements to Hardware:

Load Resistor: A 470 ohm load resistor was recommended by the lab handbook. However, a 470 ohm resistor will only require roughly 1.9W of electricity if the circuit raises the voltage to 30V. This was swapped out with a 100 ohm resistor, which draws a far more realistic 9W value.

Inductor upgrade: At high current, the typical tiny inductor is readily saturated. A specially designed toroidal inductor with a 3C90 ferrite core took its place. This substance can withstand high magnetic flux significantly more effectively. in order for the current to continue flowing.

Add Capacitors Two capacitors were connected in parallel in place of a single output capacitor. This doubles the total capacitance and halves the internal resistance, which reduces heat and makes the output voltage much smoother.

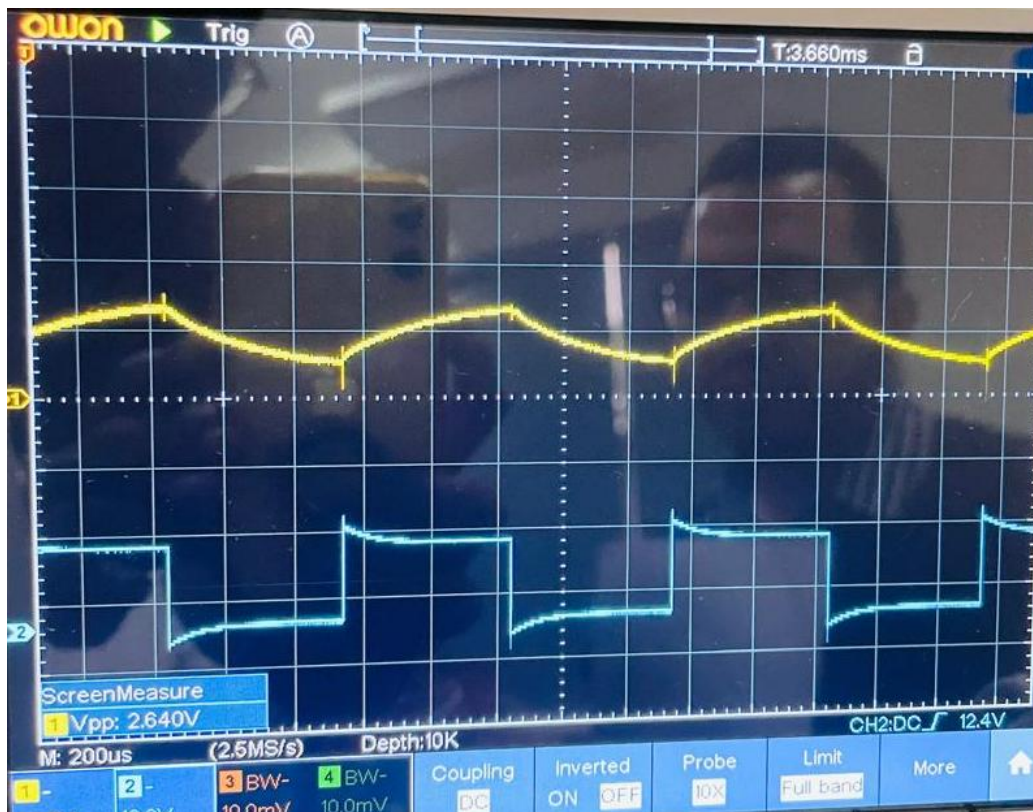


Figure 8: Scope trace of Continuous Conduction Mode (CCM).

To test the circuit, a 50% duty cycle ($D = 0.5$) was applied. The output voltage successfully doubled. Figure 8 shows the circuit running smoothly in Continuous Conduction Mode. The yellow line shows the output voltage rippling as the capacitors charge and discharge, while the blue line shows the 100 kHz switching signal.

2.3 Measurement and Protection Circuits

For the MPPT code to work, the microcontroller needs to constantly measure the voltage and current coming from the solar panel. However, the ST ARM board's ADC pins can only handle a maximum of 3.3V. If the 20V panel is connected directly, the board will break.

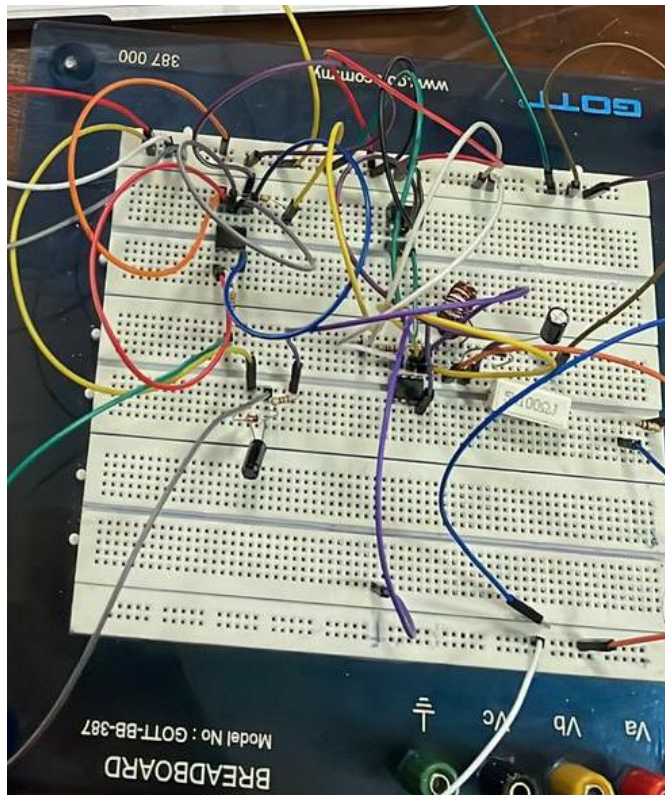


Figure 9: implementation of the MCP602 signal conditioning

Voltage and Current Sensing:

To solve the voltage issue, a voltage divider was built using a 40k-ohm and a 10k-ohm resistor.



Figure 10: Scope verification of the voltage divider.

Figure 10 proves this divider works perfectly. The 15V input from the panel is stepped down to exactly 3.0V, which is a 5:1 ratio and completely safe for the microcontroller. To make sure the microcontroller reads this voltage accurately without dropping the signal, an MCP602 op-amp was used as a buffer. A second op-amp was used to amplify the small voltage drop across a 1-ohm resistor to measure the current.

Protecting the Microcontroller:



Figure 11: Oscilloscope capture illustrating dynamic signal conditioning.

To protect the board from unexpected voltage spikes, 4.7V Zener diodes were placed in parallel with the inputs. If the voltage goes too high, the diodes safely send the extra power to the ground. Additionally, a 390-ohm resistor and a 10nF capacitor were added. Together, they create an RC low-pass filter. As seen in Figure 12, this filter rounds off sharp edges and cleans up the noisy 100 kHz switching signals, so the microcontroller gets a clean, stable reading.

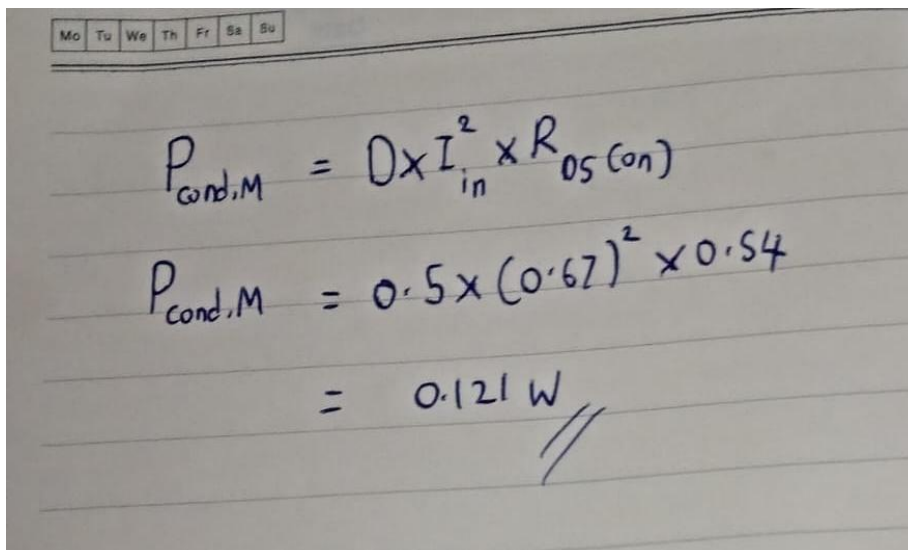
Chapter 3 – Power Loss Estimation

To observe how efficient the boost converter is, calculating the power loss as heat across the components will estimate losses by manufacturer datasheets.

For calculations, we assume the solar panel is operating at its MPP (10W), providing an input voltage of roughly 15V and an input current of 0.67A. The circuit is running at a 50% duty cycle with a switching frequency of 100 kHz. The boosted output voltage is 30V.

3.1 MOSFET Conduction Losses

When the MOSFET is turned "on," it acts like a wire, but it still has a small internal resistance called $R_{DS(on)}$. The IRF510 datasheet states the maximum $R_{DS(on)}$ is 0.54 ohms. The power lost as heat while the switch is closed is calculated as:



The image shows a handwritten calculation on lined paper. At the top, there is a small calendar header with boxes for Mo, Tu, We, Th, Fr, Sa, and Su. The calculation is as follows:

$$P_{cond,M} = D \times I_{in}^2 \times R_{DS(on)}$$
$$P_{cond,M} = 0.5 \times (0.67)^2 \times 0.54$$
$$= 0.121 \text{ W}$$

The final result, 0.121 W, is underlined with two parallel diagonal lines.

3.2 Diode Conduction Losses

When the MOSFET turns off, the current flows through the 1N4002 diode to the output. The diode has a forward voltage drop (V_f) of about 1.0V. The power lost while the diode is conducting is:

$$\begin{aligned} P_{\text{cond}, D} &= (1-D) \times I_{\text{in}} \times V_F \\ &= 0.5 \times 0.67 \times 1 \\ &= 0.335 \text{ W} \end{aligned}$$

3.3 MOSFET Switching Losses

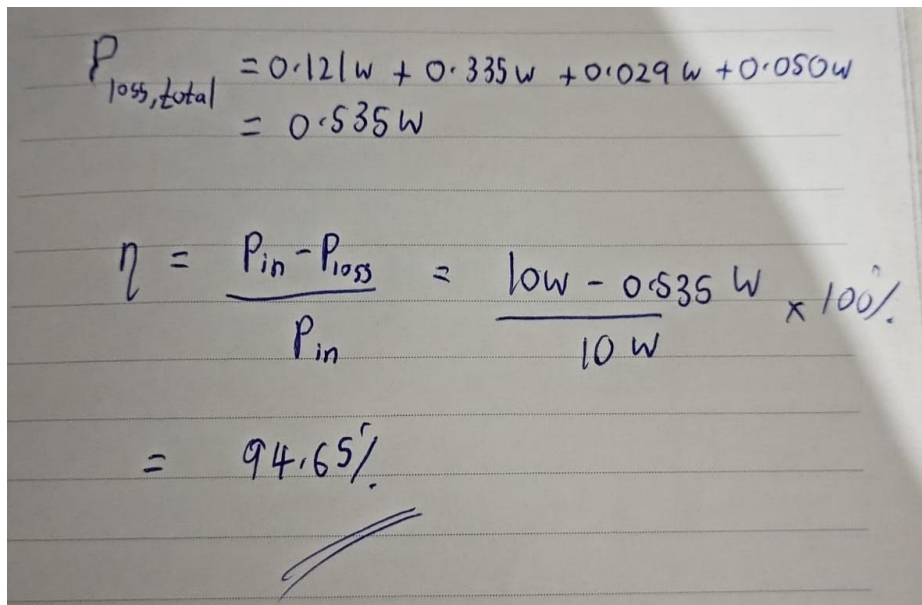
The MOSFET does not turn on and off instantly; it takes a few nanoseconds. During this transition, it experiences both high voltage and high current simultaneously, which creates switching losses. Using the IRF510 datasheet, the rise time is 16 ns, the fall time is 9.4 ns, and the output capacitance is 81 pF.

$$\begin{aligned} P_{\text{sw}, M} &= \left[\frac{1}{2} V_{\text{out}} I_{\text{in}} (t_r + t_f) f_{\text{sw}} \right] + \left[\frac{1}{2} C_{\text{oss}} V_{\text{out}}^2 f_{\text{sw}} \right] \\ &= \left[0.5 \times 30 \times 0.67 (25.4 \times 10^{-9}) \times 10^5 \right] + \\ &\quad \left[0.5 \times (81 \times 10^{-12}) \times (30)^2 \times 10^5 \right] \\ &= 0.0255 \text{ W} + 0.0036 \text{ W} = 0.029 \text{ W} \end{aligned}$$

3.4 Magnetic Core Losses

The custom 3C90 ferrite toroid inductor also experiences minor power loss due to changing magnetic fields inside the core. Based on the 3C90 material datasheet for 100 kHz operation, ferrite is highly resistive to eddy currents. We can estimate the core losses to be very small, around **0.050 W**.

Total System Efficiency



The image shows handwritten calculations on lined paper. The first part calculates the total power loss, $P_{loss, total}$, by summing four values: 0.121 W , 0.335 W , 0.029 W , and 0.050 W . The result is 0.535 W . The second part calculates the efficiency, η , using the formula $\eta = \frac{P_{in} - P_{loss}}{P_{in}}$. It substitutes 10 W for P_{in} and 0.535 W for P_{loss} , resulting in $\frac{10\text{ W} - 0.535\text{ W}}{10\text{ W}} \times 100\%$. The final result is 94.65% , which is underlined with two strokes.

$$P_{loss, total} = 0.121\text{ W} + 0.335\text{ W} + 0.029\text{ W} + 0.050\text{ W}$$
$$= 0.535\text{ W}$$
$$\eta = \frac{P_{in} - P_{loss}}{P_{in}} = \frac{10\text{ W} - 0.535\text{ W}}{10\text{ W}} \times 100\%$$
$$= 94.65\%$$

This calculation shows that the boost converter design is highly efficient. The math also proves that the fast TC4429 gate driver successfully kept the MOSFET switching losses extremely low (only 0.029W), and that the largest source of wasted energy in the circuit is actually the 1N4002 diode.

Chapter 4 – MPPT Algorithms

4.1 The Perturb and Observe (P&O) Algorithm

To actively track the Maximum Power Point (MPP), the Perturb and Observe (P&O) algorithm was implemented on the ST ARM microcontroller. This algorithm is widely utilized in industry because it is highly cost-effective, relatively simple to program, and requires minimal sensor inputs (only voltage and current) [1].

The logic works by constantly making small changes (perturbations) to the boost converter's duty cycle and observing the resulting change in power. The microcontroller reads the current and voltage via the ADC, calculates the power and compares it to the previous power reading. If the power increases, the algorithm continues to change the duty cycle in the same direction. If the power decreases, it reverses the perturbation direction [2].

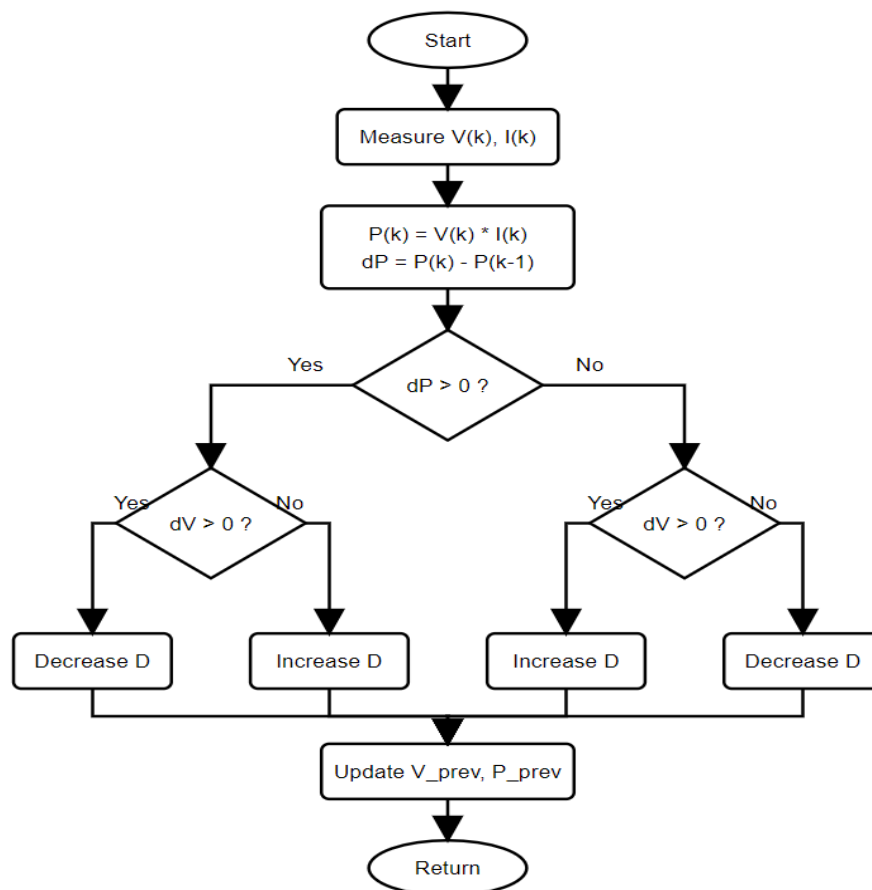


Figure 12: Flowchart of the P&O Algorithm

4.2 Alternative Algorithm 1: Incremental Conductance (INC)

While P&O is simple, research shows it constantly oscillates around the MPP under steady-state conditions and can track in the wrong direction during rapidly changing weather conditions [1]. A more advanced alternative is the Incremental Conductance (INC) algorithm, which has been proven to be more efficient in terms of fast-tracking time, higher accuracy, and lower steady-state oscillation [2].

INC is based on the mathematical principle that at the maximum point the derivative of power respect to voltage is exactly zero. Since $P = V \cdot I$, at the MPP it can be written as I/V is equal to $\Delta I / \Delta V$

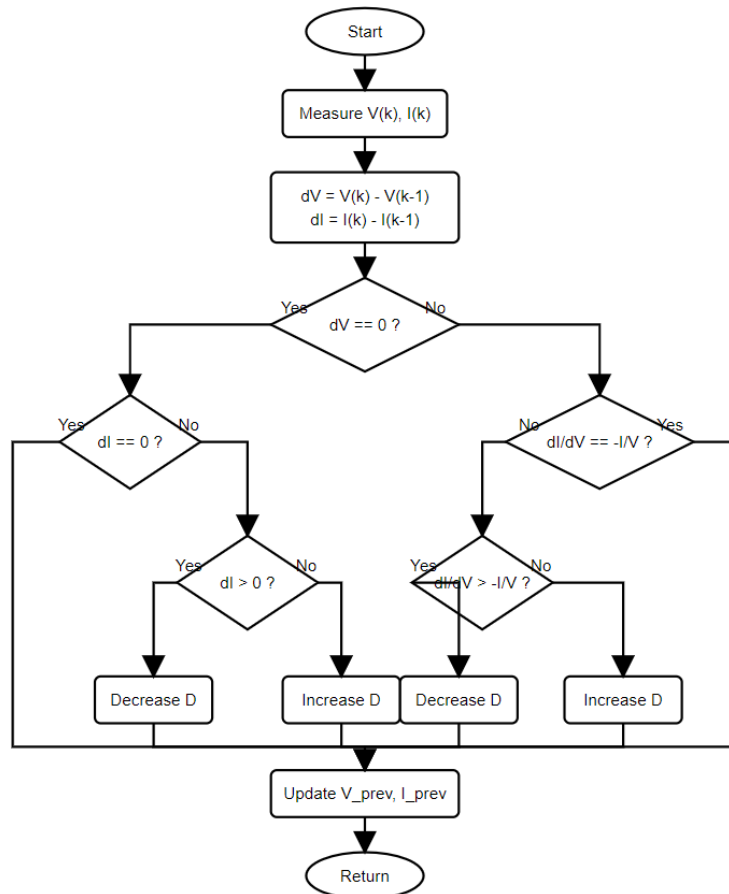


Figure 13: Flowchart of the Incremental Conductance Algorithm.

Pseudo-Code for Incremental Conductance:

Read V_{new} , I_{new}

Calculate $dV = V_{new} - V_{old}$

Calculate $dI = I_{new} - I_{old}$

IF $dV = 0$:

IF $dI = 0$:

Do nothing (Already at MPP)

ELSE IF $dI > 0$:

Decrease Duty Cycle

ELSE:

Increase Duty Cycle

ELSE:

IF $(dI / dV) = -(I_{new} / V_{new})$:

Do nothing (Already at MPP)

ELSE IF $(dI / dV) > -(I_{new} / V_{new})$:

Decrease Duty Cycle

ELSE:

Increase Duty Cycle

Update $V_{old} = V_{new}$, $I_{old} = I_{new}$

4.3 Alternative Algorithm 2: Fractional Open Circuit Voltage (FOCV)

This algorithm utilizes the approximately linear relationship between a solar panel's Maximum Power Point voltage (V_{MPP}) and its open-circuit voltage (V_{OC}) [3].

Empirical data dictates that V_{MPP} is usually a fixed fraction of V_{OC} , typically ranging between 0.71 to 0.78 depending on the specific photovoltaic material and environmental temperature [4]. The algorithm works by momentarily disconnecting the panel from the load to measure the true V_{OC} , multiplying it by a constant fraction (k), and then setting the boost converter to hold that target voltage [3].

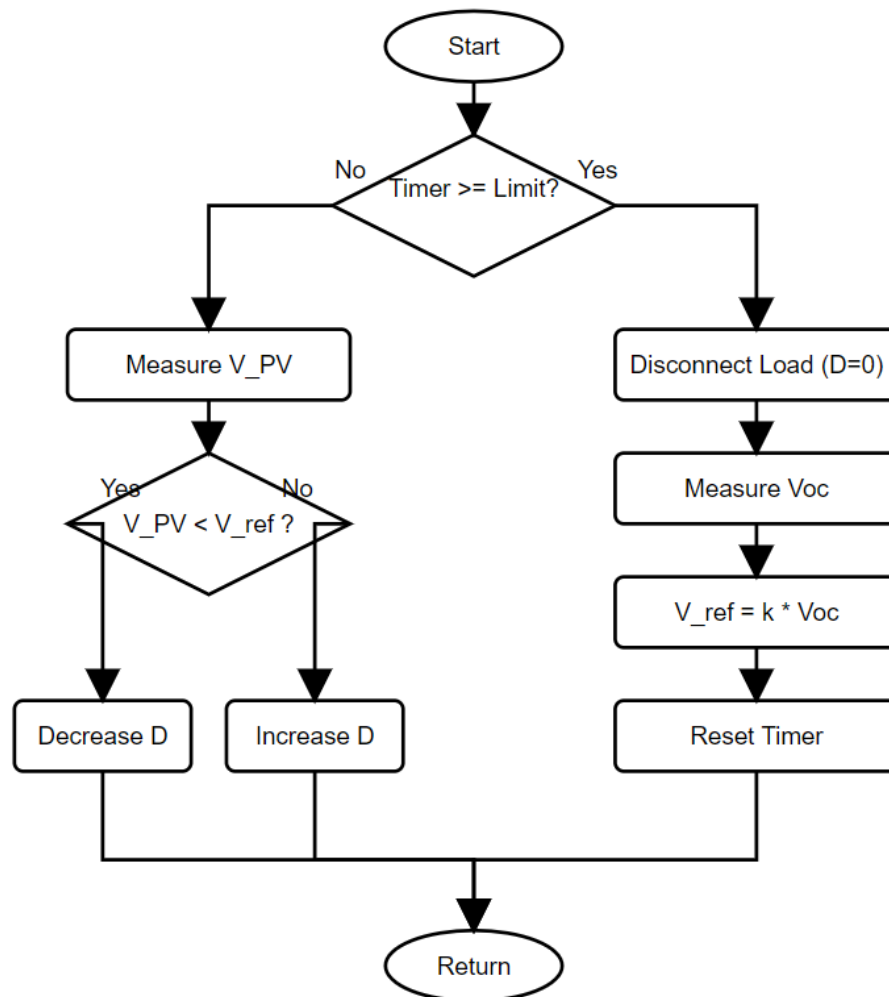


Figure 14 : Flowchart of the Fractional Open Circuit Voltage Algorithm

Pseudo-Code for Fractional Open Circuit Voltage:

Define constant k (e.g., 0.75)

Define timer interval

IF Timer = Interval limit:

Disconnect boost converter (Duty Cycle = 0)

Wait for voltage to stabilize

Read V_new (This is now V_oc)

Calculate $V_{target} = k * V_{new}$

Reset Timer

ELSE:

Read V_panel

IF $V_{panel} < V_{target}$:

Decrease Duty Cycle

ELSE IF $V_{panel} > V_{target}$:

Increase Duty Cycle

4.4 Comparative Algorithmic Analysis

To determine the optimal use case for each algorithm, they are evaluated across four primary engineering metrics: complexity, reliability, stability, and reaction time [1], [4].

Table 1: Comparative Algorithmic Analysis

Metric	Perturb and Observe (P&O)	Incremental Conductance (INC)	Fractional Open Circuit (FOCV)
Complexity	Low. Very easy to program using basic math [1].	High. Requires complex division and edge-case handling (e.g., division by zero) [2].	Very Low. Requires only a voltage sensor and simple multiplication [3].
Reliability	Moderate. Fails or tracks incorrectly during fast changes in solar irradiance [1].	High. Highly accurate even when sunlight changes rapidly [2].	Low. It only estimates the MPP based on a constant, rather than finding the true peak [4].
Stability	Low. Constantly oscillates back and forth around the MPP, wasting small amounts of power [1].	High. Once it finds the exact MPP ($dP/dV = 0$), it stops adjusting and holds steady [2].	Moderate. It holds a steady voltage but requires physically disconnecting power periodically to measure V_{OC} [3].
Reaction Time	Moderate. Depends heavily on the size of the perturbation step chosen [2].	Fast. Mathematically calculates exactly which side of the curve it is on instantly [1].	Slow. Temporarily stops power generation completely to take new measurements [4].

References

- [1] M. E. Ahmad *et al.*, "A Comparative Study of Perturb and Observe (P&O) and Incremental Conductance (INC) PV MPPT Techniques," *Engineering and Technology Journal*, vol. 40, no. 02, pp. 376-385, 2022.
- [2] M. Bouksaim, M. Mekhfioui, and M. N. Srifi, "A Comprehensive Decade-Long Review of Advanced MPPT Algorithms for Enhanced Photovoltaic Efficiency," *Solar*, 2025.
- [3] "An Enhanced Fractional Open Circuit Voltage MPPT Method for Rapid and Precise MPP Tracking in Standalone Photovoltaic Systems," *IEEE Access*, 2025.
- [4] "Improved Fractional Open Circuit Voltage MPPT Methods for PV Systems," *MDPI Electronics*, vol. 8, no. 3, 2019.

Appendices

Appendix A: Statement of Artificial Intelligence Usage In accordance with the University's academic integrity guidelines, I declare the use of AI-assisted technologies (specifically Google Gemini) during the preparation of this laboratory report.

How AI was utilized:

- **Proofreading and Tone:** AI was used to review my original drafted text for grammatical accuracy and to ensure a formal, professional engineering tone.
- **Formatting and Structure:** AI assisted in organizing my experimental data, calculations, and laboratory photographs into the required rubric structure.
- **Code Formatting:** AI was used to format and condense my original C-code logic for the Perturb and Observe algorithm into a readable format for the appendix.

All experimental data, oscilloscope waveforms, physical hardware iterations (e.g., component values, breadboard assemblies), mathematical power loss calculations, and core engineering conclusions are entirely my own original work, derived directly from the laboratory session.

Appendix B: Microcontroller C-Code

C

```
#include <stm32f3xx.h>

#include <math.h>

#include <stdio.h>

#include "adc.h"

#include "leds.h"

#include "usart.h"

#include "systick_delay.h"

#include "PLL_Config.h"

#include "pwm.h"


// Hardware factors

#define V_SCALE 5.0f // 40k and 10k divider ratio

#define I_SCALE 1.0f // 1 ohm shunt, no amp gain

#define V_REF 3.3f


// reduce switching noise

float get_adc_avg(int channel) {

    long sum = 0;

    for(int i = 0; i < 8; i++) {

        sum += (channel == 0) ? convert_PA0 : convert_PA1;

        delay_nms(1);

    }

    return (float)sum / 8.0f;
```

```
}
```

```
float read_voltage() {  
    return (get_adc_avg(0) / 4095.0f) * V_REF * V_SCALE;  
}
```

```
float read_current() {  
    return (get_adc_avg(1) / 4095.0f) * V_REF / I_SCALE;  
}
```

```
int main(void) {  
    // Init system  
    PLL_Config();  
    SystemCoreClockUpdate();  
    SysTick_Init();  
    init_usart(115200);  
    ADC_Init();  
    init_led();  
    init_pwm(100000); // 100 kHz switching
```

```
    float duty = 50.0f;  
    float P_prev = 0.0f;  
    float V_prev = 0.0f;  
    float perturb = 0.5f;  
    char msg[64];
```

```

output_pwm(duty);

delay_nms(500); // Let hardware settle


while(1) {
    led_on();


    float V = read_voltage();
    float I = read_current();
    float P = V * I;


    float dP = P - P_prev;
    float dV = V - V_prev;


    // P&O Logic with 5mW dead zone to stop jitter at MPP
    if (fabsf(dP) > 0.005f) {
        if (dP > 0) {
            perturb = (dV >= 0) ? 0.5f : -0.5f;
        } else {
            perturb = (dV >= 0) ? -0.5f : 0.5f;
        }
    }


    duty += perturb;


    // Hard limits for safety
    if(duty > 90.0f) duty = 90.0f;
}

```



```

if(duty < 10.0f) duty = 10.0f;

output_pwm(duty);

P_prev = P;
V_prev = V;

// Update indicator LEDs
if (fabsf(dP) <= 0.005f) {
    orange_led_on(); green_led_off(); red_led_off(); // MPP Found
} else if (dP > 0) {
    green_led_on(); orange_led_off(); red_led_off(); // Tracking
} else {
    green_led_off(); orange_led_off();
}

if (duty >= 90.0f || duty <= 10.0f) {
    red_led_on(); // Duty limit reached
} else {
    red_led_off();
}

// Output
sprintf(msg, "V=%.2f I=%.3f P=%.3f D=%.1f\r\n", V, I, P, duty);
print_terminal(msg);

```

```
    led_off();  
    delay_nms(100);  
}  
}
```